🎓 LearnHub

Online Learning Platform

System Design — Final Examination Project Documentation

Submitted by: _____

Roll No: _____

Course: System Design

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1. Problem Statement

Learners lack a single, structured platform to discover quality courses, enroll, follow an organized curriculum, and validate their understanding through assessments. Instructors lack a simple way to publish structured course content and measure learner performance. A scalable system is needed that cleanly separates concerns — authentication, course management, enrollment, and assessment — so each can evolve and scale independently.

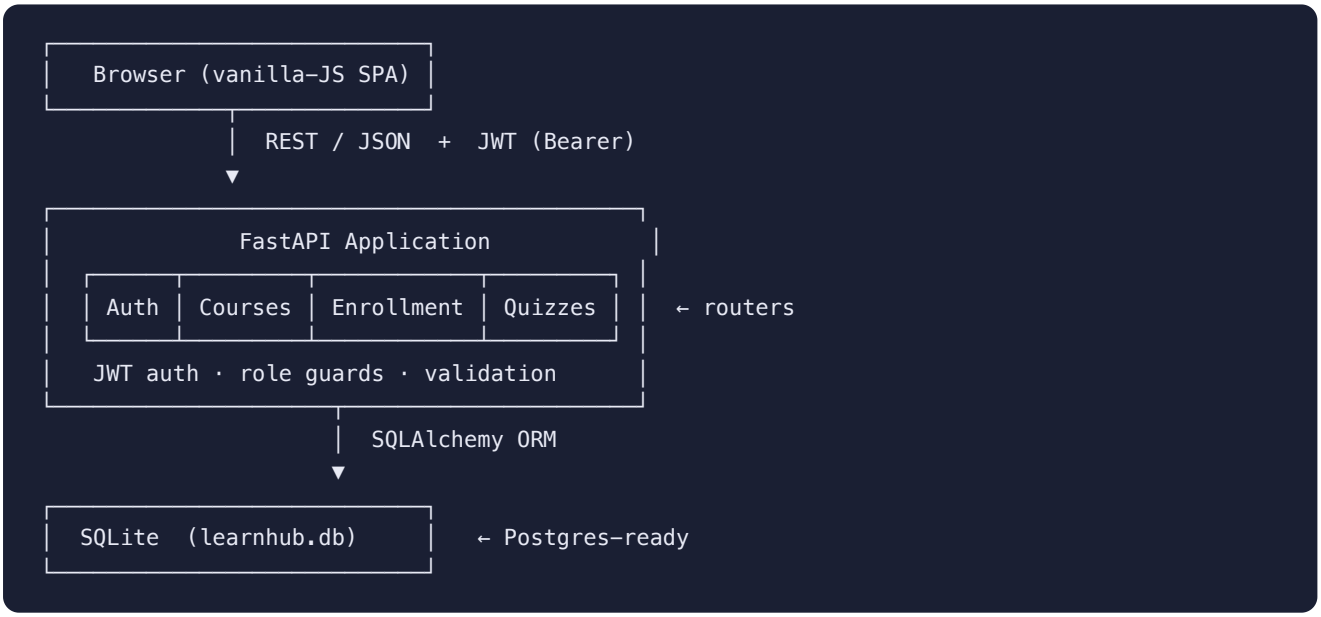
2. Proposed Solution

LearnHub is a modular web platform built on a **FastAPI** backend exposing a REST API, backed by a relational database via the **SQLAlchemy** ORM, with a lightweight JavaScript frontend. Functionality is split into four independent modules:

- **Auth** — JWT-based authentication with role-based access (learner, instructor, admin).
- **Courses** — instructors create courses organized into sections and lessons; learners browse, search, and filter.
- **Enrollment** — learners enroll in courses and track lesson-level progress.
- **Quizzes** — instructors author quizzes; the server auto-grades submissions and enforces attempt limits.

The modular router design mirrors a microservice-style separation while remaining a single deployable app — each module can later be extracted into its own service.

3. System Architecture



Request flow: the client sends JSON with a JWT in the `Authorization` header → FastAPI validates the token and the request body (Pydantic) → the matching router enforces role/ownership rules → SQLAlchemy reads/writes the database → a validated JSON response is returned. Detailed architecture, sequence, and ER diagrams are maintained in the `learnhubdiagrams/` folder.

4. Module Description

Module	Responsibility	Key Endpoints
Auth	Registration, login, JWT issuance, role enforcement, password hashing (PBKDF2).	POST /api/auth/register POST /api/auth/login GET /api/auth/me
Courses	Course/section/lesson CRUD, public catalog with search & category filter, ownership checks.	GET /api/courses GET /api/courses/{id} POST /api/courses
Enrollment	Enroll learners, list "My Learning", update progress percentage.	POST /api/enrollments/{cid} GET /api/enrollments PATCH …/progress
Quizzes	Author quizzes/questions, serve quizzes without leaking answers, auto-grade, enforce max attempts.	POST /api/courses/{id}/quizzes GET /api/quizzes/{id} POST /api/quizzes/{id}/submit

5. Database Design

Ten relational tables model the domain. Primary relationships:

- `users` 1—N `courses` (an instructor owns many courses)
- `courses` 1—N `sections` 1—N `lessons` 1—N `resources`
- `users` N—N `courses` via `enrollments`
- `courses` 1—N `quizzes` 1—N `questions`
- `quiz_attempts` 1—N `attempt_answers` (one row per answered question)

Table	Important Columns
users	id, email (unique), password_hash, full_name, role, is_verified, created_at
courses	id, instructor_id (FK), title, description, category, price, is_free, rating_avg, review_count
sections	id, course_id (FK), title, position
lessons	id, section_id (FK), title, video_url, duration, is_free_preview, position
resources	id, lesson_id (FK), title, url, type
enrollments	id, user_id (FK), course_id (FK), progress, enrolled_at
quizzes	id, course_id (FK), title, pass_mark, max_attempts, order_index
questions	id, quiz_id (FK), type, prompt, options (JSON), correct_answer, position
quiz_attempts	id, quiz_id (FK), enrollment_id (FK), user_id (FK), score, passed, attempt_number
attempt_answers	id, attempt_id (FK), question_id (FK), answer_given, is_correct

6. Technology Stack

Layer	Technology	Why
Language	Python 3.11+	Required; rich ecosystem
API framework	FastAPI	Async, auto OpenAPI docs, Pydantic validation
ORM	SQLAlchemy 2.0	DB-agnostic models; easy SQLite→Postgres
Database	SQLite	Zero-setup; file-based; production swaps to PostgreSQL
Auth	PyJWT + hashlib PBKDF2	Stateless tokens; no native build deps
Server	Uvicorn (ASGI)	Fast async server
Frontend	HTML/CSS/JS	No build step; easy to run and demo

7. Implementation Details

Authentication. Passwords are hashed with PBKDF2-HMAC-SHA256 (200k iterations, per-user salt) using the standard library. On login the server issues a JWT carrying the user id and a 24-hour expiry. Protected routes depend on `get_current_user`; role-restricted routes use a `require_role(...)` dependency factory.

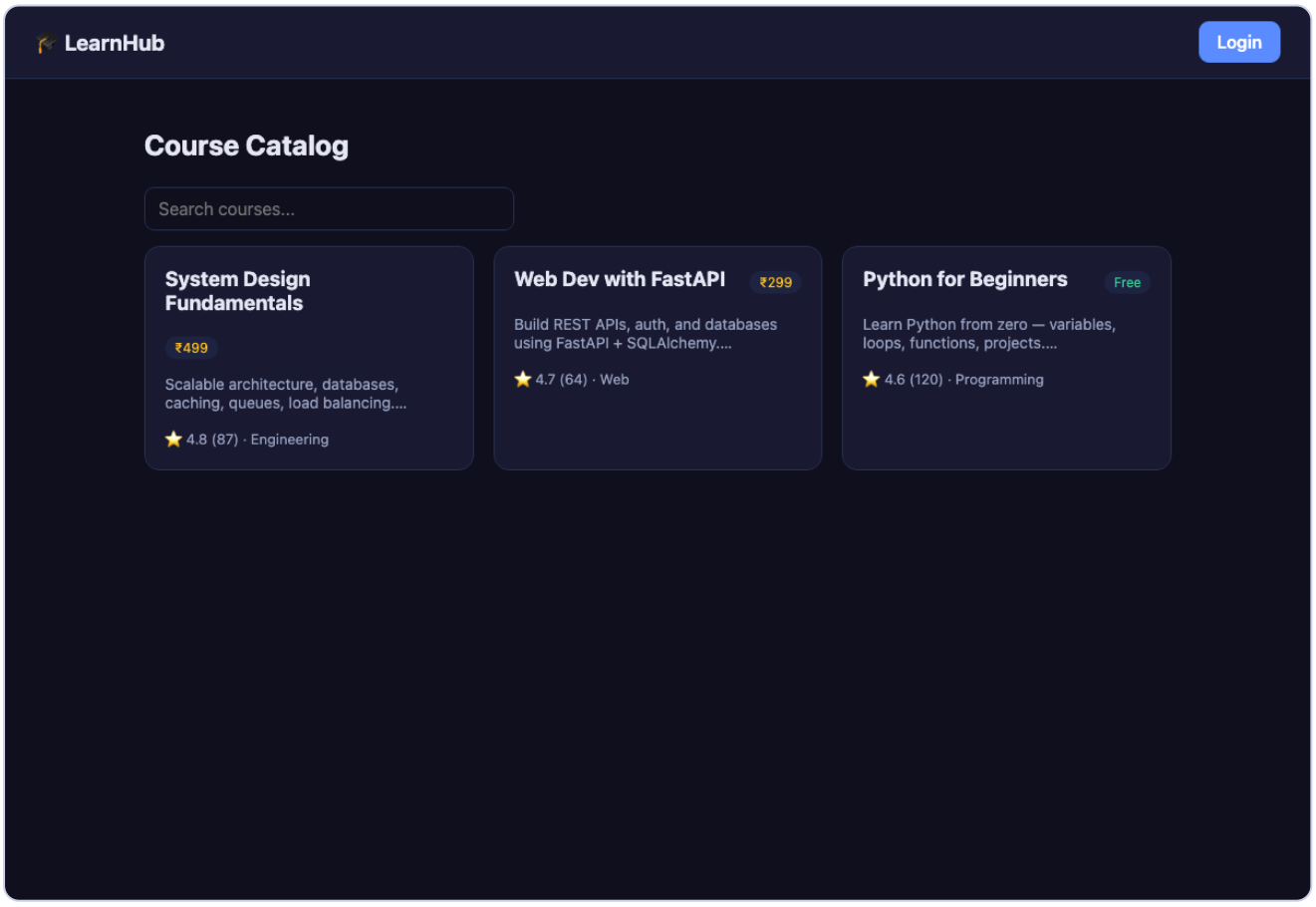
Quiz auto-grading. The quiz-fetch endpoint deliberately strips `correct_answer` so it can never be read from the browser. On submit, the server compares each answer case-insensitively, computes a percentage score, marks the attempt passed/failed against the quiz `pass_mark`, and persists every answer for review. `max_attempts` is enforced server-side.

Validation & safety. Every request/response is typed with Pydantic schemas, giving automatic 422 errors on bad input and a self-documenting OpenAPI spec. Ownership checks prevent one instructor from editing another's course.

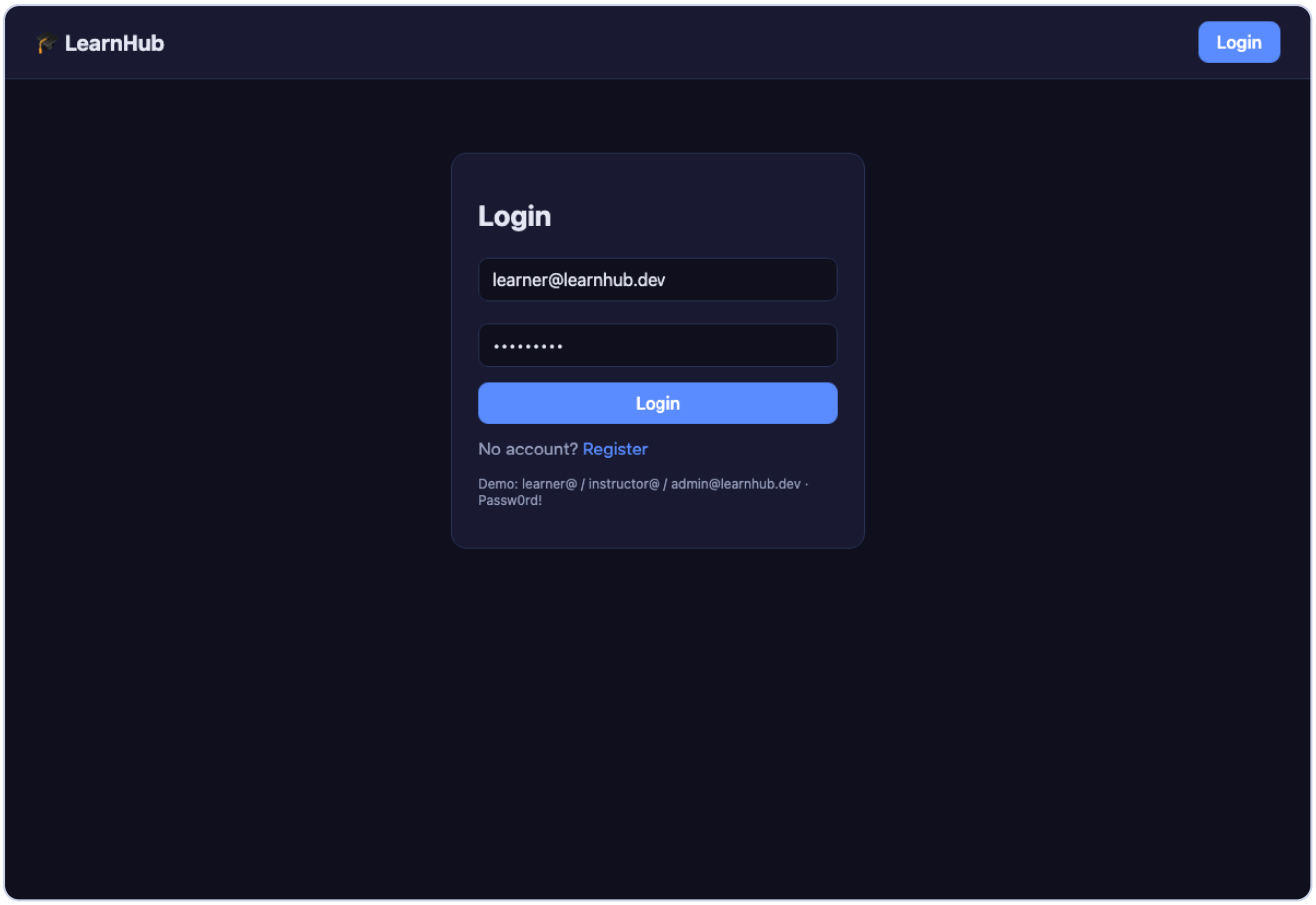
```
POST /api/quizzes/1/submit          # all-correct submission
→ { "score": 100.0, "passed": true, "correct": 3, "total": 3, "attempt_number": 1 }

POST /api/courses   (as learner)    # role guard blocks non-instructors
→ 403  { "detail": "Requires role: instructor, admin" }
```

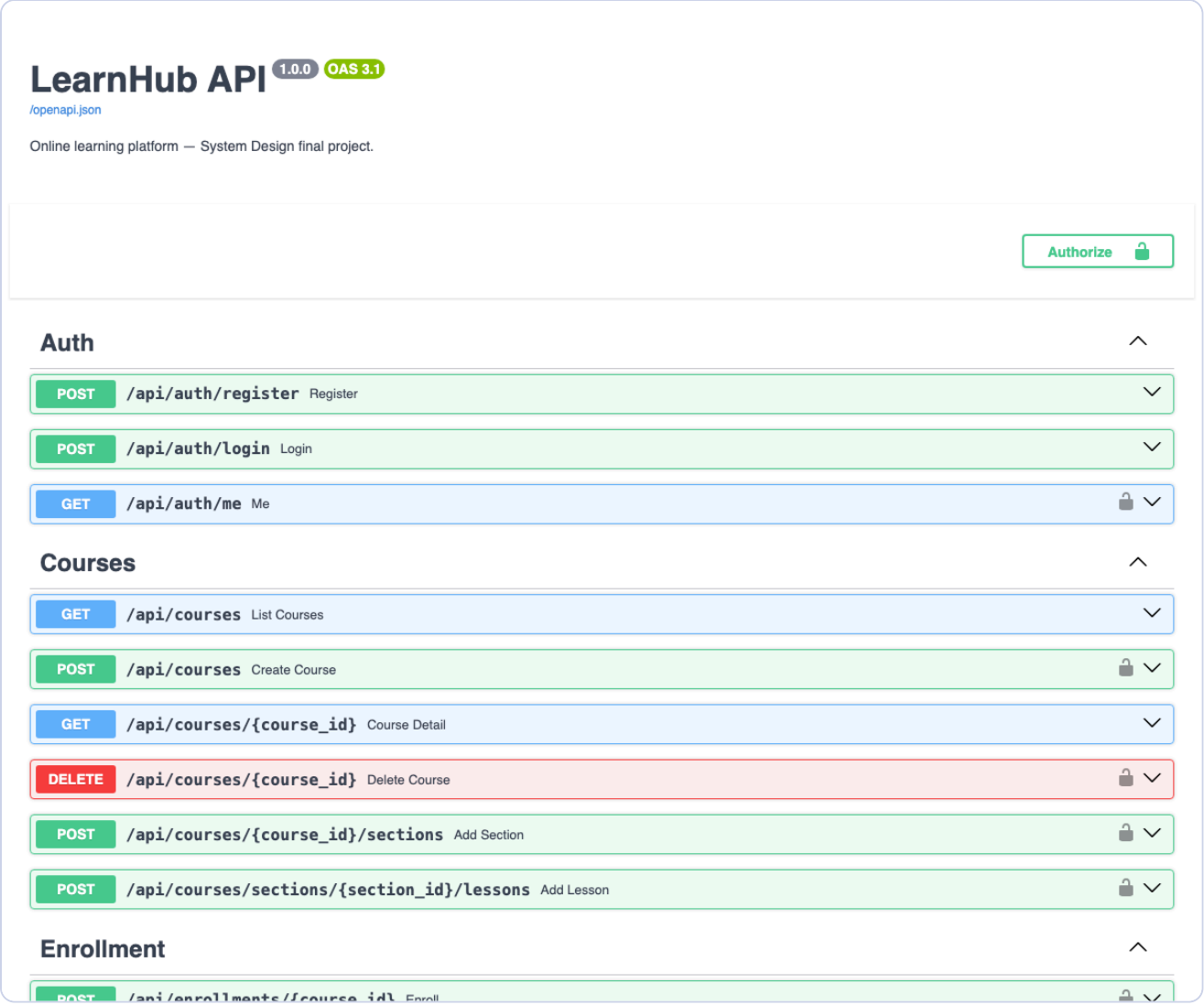
8. Screenshots



Course catalog — search, ratings, free/paid badges.



Login screen with demo accounts.



Auto-generated Swagger API documentation at /docs.

Additional screens (My Learning dashboard, quiz attempt & result) can be captured after logging in locally with the demo accounts.

9. Future Scope

- Razorpay payment gateway with webhook-confirmed paid enrollment.
- Media uploads to S3/CDN with presigned URLs and video streaming.
- Migration to PostgreSQL with Alembic migrations; full-text search ranking.
- ML-based recommendation engine (scikit-learn) for a personalized feed.
- Celery task queue for emails, certificate generation, and analytics.
- Email/OTP verification, password reset, and instructor payout management.